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Studierichting: Informatica
Oefeningenreeks 5D nr. 6.1

$y^2 = 6x$ en $x^2 = 6y$

snijpunten

$$\sqrt{6x} = \frac{x^2}{6} \Leftrightarrow 6x = \frac{x^4}{36} \Leftrightarrow 216x = x^4 \Leftrightarrow x(x^3 - 216) \Leftrightarrow x = 0 \vee x = 6$$

$$\int_0^6 \left( \sqrt{6x} - \frac{x^2}{6} \right) dx$$

$$= \int_0^6 \left( \sqrt{6}\sqrt{x} \right) dx - \int_0^6 \frac{x^2}{6} dx$$

$$\left[ \frac{2}{3}\sqrt{6}x^{\frac{3}{2}} \right]_0^6 - \left[ \frac{x^3}{18} \right]_0^6$$

$$= (24 - 0) - (12 - 0) = 12$$

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# References